

Multimodal Deep Learning Classifier for Primary Open Angle Glaucoma Diagnosis Using Wide-Field Optic Nerve Head Cube Scans in Eyes With and Without High Myopia

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Précis: An optical coherence tomography (OCT)-based multimodal deep learning (DL) classification model, including texture information, is introduced that outperforms single-modal models and multimodal models without texture information for glaucoma diagnosis in eyes with and without high myopia.

Background/Aims: To evaluate the diagnostic accuracy of a multimodal DL classifier using wide OCT optic nerve head cube scans in eyes with and without axial high myopia.

Materials and Methods: Three hundred seventy-one primary open angle glaucoma (POAG) eyes and 86 healthy eyes, all without axial high myopia [axial length (AL) ≤ 26 mm] and 92 POAG eyes and 44 healthy eyes, all with axial high myopia (AL > 26 mm) were included. The multimodal DL classifier combined features of 3 individual VGG-16 models: (1) texture-based en face image, (2) retinal nerve fiber layer (RNFL) thickness map image, and (3) confocal scanning laser ophthalmoscope (cSLO) image. Age, AL, and disc area adjusted area under the receiver operating curves were used to compare model accuracy.

Results: Adjusted area under the receiver operating curve for the multimodal DL model was 0.91 (95% CI = 0.87, 0.95). This value was significantly higher than the values of individual models [0.83 (0.79, 0.86) for texture-based en face image; 0.84 (0.81, 0.87) for

RNFL thickness map; and 0.68 (0.61, 0.74) for cSLO image; all $P \leq 0.05$]. Using only highly myopic eyes, the multimodal DL model showed significantly higher diagnostic accuracy [0.89 (0.86, 0.92)] compared with texture en face image [0.83 (0.78, 0.85)], RNFL [0.85 (0.81, 0.86)] and cSLO image models [0.69 (0.63, 0.76)] (all $P \leq 0.05$).

Conclusions: Combining OCT-based RNFL thickness maps with texture-based en face images showed a better ability to discriminate between healthy and POAG than thickness maps alone, particularly in high axial myopic eyes.

Key Words: glaucoma, myopia, optical coherence tomography, deep learning, multimodal, classification

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Epidemiological evidence indicates that myopia is projected to impact 50% of the world's population by 2050, with a strong association between myopia and glaucoma.¹ Studies show that individuals with myopia have a 2.5-fold increased risk of developing glaucoma compared with nonmyopic individuals, whereas high myopes face an even greater risk of 5–6 times for developing glaucomatous optic neuropathy.^{2–4} This underscores the significant correlation between myopia and glaucoma, highlighting the need for awareness and proactive management of myopia to mitigate the risk of glaucoma development.

Over the past 2 decades, machine learning methods have been applied to various imaging and visual function measurements, and have shown improved classification of healthy, primary open angle glaucoma (POAG) suspect, and POAG eyes compared with instrument-based parameters in a large number of studies that have been summarized in several review articles.^{5–11} In a recent study using single-modal classifiers, we showed that ResNet50 deep learning (DL) models were trained and tested on the retinal nerve fiber layer (RNFL) en face. Spectralis optical coherence tomography (OCT) images outperformed DL models based on RNFL thickness maps and confocal scanning laser ophthalmoscopy images and mean RNFL thickness and circumpapillary (ie circle scan) RNFL thickness measurements for discriminating between eyes with repeatable glaucomatous visual field (VF) defects [mean deviation (MD) = -5.7 dB] and eyes without GFVDs (MD = -0.04 dB).¹² A similar pattern of results [with reduced area under the receiver operating curve (AUROCs)] was observed in a subset analysis of eyes with mild glaucomatous VF defects (MD < -6 dB). Although the previous study did not

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use the specific texture analysis for the RNFL en face images, the better performance of the RNFL en face images may have been due, in part, to texture properties reflected in the en face image.

Recently, a novel OCT texture-based macular en face image analysis method [called San Diego Automated Layer Segmentation Algorithm (SALSA)-Texture] requiring segmentation of only one retinal surface (the internal limiting membrane) was shown to have higher diagnostic accuracy for discriminating between highly myopic healthy eyes and highly myopic glaucoma eyes as compared with standard macula layer thickness measurements.¹³

Evidence over decades has indicated that combining single-modal models (ie, models using a single measurement technique) for POAG detection into hybrid models can improve diagnostic accuracy. For instance, over 15 years ago, Shah et al¹⁴ combined imaging measurements (confocal scanning laser ophthalmoscopy, time-domain OCT, and scanning laser polarimetry) with visual function measurements (standard automated perimetry, frequency-doubling technology perimetry, and short-wave automated perimetry) and determined that adding functional information to structural measurements improved classification of healthy and POAG eyes. Subsequently, similar multimodal model improvement has been reported.^{15–24}

Despite the improvements in classification performance for healthy versus glaucomatous eyes provided by multimodal models, en face images, and machine learning analyses, there is concern that imaging of the optic disc is not ideal for detecting glaucoma in highly myopic eyes. In particular, the increased axial length (AL) significantly reduces global measured circumpapillary RNFL thickness and results in a different regional arrangement of the retinal nerve fibers. These differences in sectoral values may result in myopic eyes being incorrectly classified as outside normal limits by imaging instruments used for glaucoma detection.^{25–28}

The objective of the current study is to determine whether combining optic nerve head (ONH) and partial macula measurements, including texture measurements from wide-field en face images, in multimodal models using machine learning analyses can improve healthy versus glaucoma classification performance in general and in highly myopic eyes in particular.

MATERIALS AND METHODS

This study was carried out in accordance with the principles of the Declaration of Helsinki. The University of California San Diego (UCSD) Institutional Review Board and Human Subjects Committee approved all protocols and methods. This study was registered [at <http://clinicaltrials.gov> (no. NCT00221923)] on September 14, 2005. Participants included in this cross-sectional observational study were recruited from the Diagnostic Innovations in Glaucoma Study (DIGS). Details of the DIGS protocol have been described.²⁹

Participants

Available data included demographics, ocular and medical history, best-corrected visual acuity, slit-lamp biomicroscopy, intraocular pressure (IOP) measured by Goldmann applanation tonometry, gonioscopy, central corneal thickness measured by ultrasound pachymetry (DGH Technology, Inc, Exton), mydriatic fundus examination,

simultaneous stereophotography of the optic disc, VF testing using standard automated perimetry (Humphrey Field Analyzer; 24-2 with Swedish Interactive Thresholding Algorithm standard; Carl Zeiss Meditec), and Spectralis OCT2 imaging (HEYEX version 6.10; Heidelberg Engineering GmbH).

Overall inclusion criteria were age 18 years or older, open anterior chamber angles upon gonioscopy, and a best-corrected visual acuity of 20/40 or better at study entry. Patients were excluded if they had a history of ocular or systemic diseases that could affect the VF or optic nerve including peripapillary staphylomas. Unreliable VFs and poor-quality OCT scans were excluded.

All VFs were evaluated by UC San Diego VF Assessment Center personnel based on a standardized protocol.²⁹ Perimetries with fixation losses of more than 33%, false-positive errors of more than 33%, and glaucoma hemifield test (GHT) classification of “abnormally high sensitivity” were deemed unreliable and were excluded.

POAG was defined based on the DIGS conventional standard of glaucomatous VF loss (pattern SD < 95% confidence limits or a GHT outside of normal limits) and corresponding optic disc/parapapillary damage.²⁹ Stereo photograph-based glaucomatous damage was defined as focal or diffuse narrowing of the neuroretinal rim, and/or detection of RNFL defects characteristic of glaucoma based on a masked assessment by 2 trained observers. Two experts (J.R. and C.B.) graded photographs after high myopia optic disc grading training with a senior consultant (J.B.J.) with expertise in myopia and glaucoma. Stereo photograph-based optic disc damage was defined by consensus between both graders. In case of disagreement, diagnosis was defined by adjudication by the senior consultant.

Healthy individuals were required to have an IOP of ≤ 21 mm Hg with no history of elevated IOP, normal-appearing optic discs, intact neuroretinal rims and peripapillary RNFLs and a minimum of 2 reliable and normal VFs with pattern SD within 95% confidence limits and a GHT result within normal limits oculus uterque.

The data set included data from both eyes with and without high axial myopia defined as AL > 26.0 mm as described further. Data are available from the corresponding author upon reasonable request.

High Axial Length Myopia Definition

High axial myopia eyes required an AL of > 26.0 mm. Myopia was defined by AL because axial elongation can lead to morphologic changes in the optic disc and the fundus.²⁷ Such changes are not reflected when myopia is defined by refractive error only. For example, cataract surgery or refractive procedures can result in a refractive shift in eyes that are axially elongated but such eyes would no longer be classified as (highly) myopic by refractive error.

Optical Coherence Tomography

The Spectralis OCT2 (HEYEX version 6.10; Heidelberg Engineering GmbH) was used for OCT imaging. This instrument uses an 870 nm central wavelength at an 85 kHz A-scan rate. We used a custom wide-field ONH OCT cube scan of 30 degrees $\times$ 25 degrees (8.7 $\times$ 7.3 mm) centered on the optic disc composed of 121 horizontal B-scans composed of 768 A-scans each.¹³ B-scans were 60 μ m apart with a lateral resolution was 5.64 μ m/pixel; axial resolution was 3.87 μ m/pixel and a frame rate was 10 per B-scan. Scan time ranged from $\sim$ 1 to 5 seconds. Images were

electronically transferred to the UC San Diego Imaging Data Evaluation and Analysis Center for quality assessment. Quality review of Spectralis images required well-centered scans with no motion artifacts and signal strength > 15 dB. All images were deemed acceptable quality for use based on standardized qualitative assessment according to Imaging Data Evaluation and Analysis Center protocol. ONH scanning laser ophthalmoscope (SLO) image quality was manually confirmed by one author (A.B.).

Retinal Layer Segmentation

Raw 3-dimensional wide ONH SD-OCT volume scans were exported to a mathematical computing software (MATLAB; MathWorks). The SALSA-deep was used to automatically detect the inner limiting membrane (ILM) and to segment the RNFL.

Texture En Face Image

Texture can be characterized as a visual pattern that reflects the spatial arrangement of pixel intensities of an image. Texture analysis captures the granularity and repetitive patterns of object surfaces. In the case of OCT images, each retinal layer has a unique texture that can be visually distinguished. We recently proposed a new texture descriptor called SALSA-texture, which is robust to the local variations in tissue reflectivity caused by varying illumination.¹³ In brief, a slab 70 μm thick from the ILM was created. For each pixel of a given B-scan, a local feature descriptor computed as the mean difference between the average of a 9×9 normalized difference of the Gaussian filter and the pixel itself was computed. Homogeneous-bin normalization was then used to increase the robustness of the local feature descriptors.³⁰

Single-modal Convolutional Neural Network (CNN)

In this work, VGG-16³¹ DL models were used. VGG-16 has 13 convolutional layers, followed by rectification and pooling layers, and 3 fully connected layers. All

convolutional layers used small 3×3 filters and the network performed only 2×2 pooling.

Specifically, we fine-tuned 3 VGG-16 models for glaucoma classification using (1) texture-based en face images, (2) RNFL thickness map images, and (3) SLO images. All 3 CNN architectures had weights pretrained on the ImageNet data set using TensorFlow 1.15.0 as a back end for Keras 2.1.6 on Python 3.6.5 for Linux. The VGG-16 network weights were frozen in the first 4 convolution blocks. Weights in the final convolutional block were allowed to update during fine-tuning. An Adam optimizer with a learning rate of 1×10^{-5} and a dropout rate of 50% was used to train the network.

Multimodal CNN

Performance of the individual texture-based en face, RNFL thickness-based, and SLO image-based CNNs were compared with a multimodal (ie, multi-input) CNN architecture using all available imaging measurements from each image type to maximize the available data and to hypothetically improve performance. Three parallel VGG-16 subarchitectures with similar layers as the single-layer architectures described previously composed this architecture. Before applying the first fully connected layer, a flattened layer was applied to the extracted features from (1) texture-based en face images, (2) RNFL thickness map images, and (3) SLO images after which a concatenation was applied. The architecture of the multimodal VGG-16 network is shown in Figure 1.

Training and Evaluation

Five-fold cross-validation was used to provide out-of-sample predictions for DL CNN models to avoid over-optimistic estimates of classification accuracy. Both healthy and glaucoma eyes were randomly divided at the patient level into 5 subsets. For each model, we used 4 subsets to train the model and used the fifth subset to assess model performance. This sequence was repeated 5 times, with each subset serving as the test set one time so that each tested eye

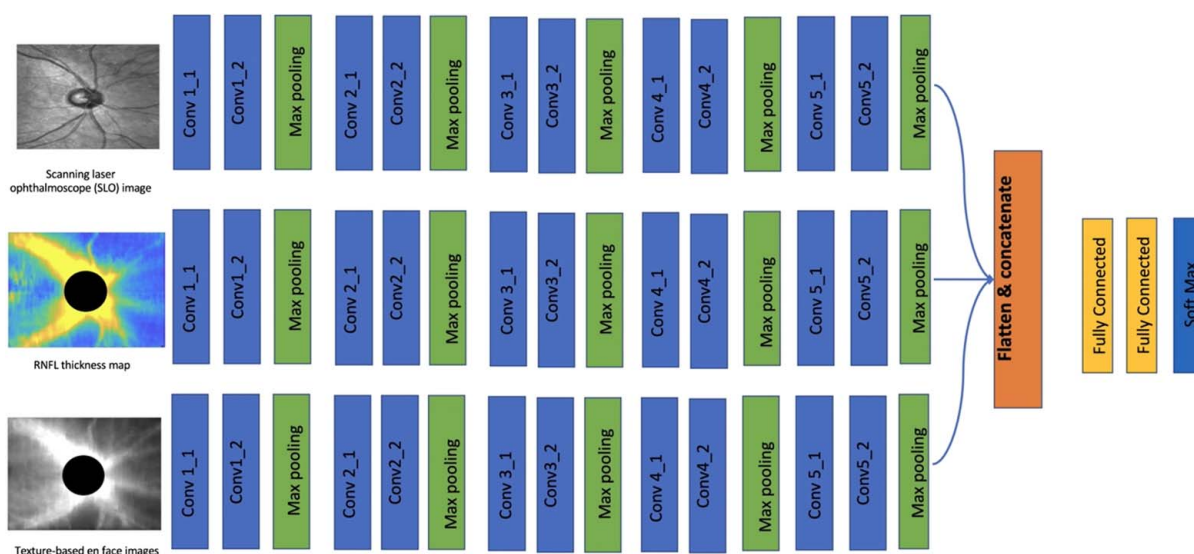


FIGURE 1. Architecture of the multimodal VGG-16 network and example of the wide-field scanning laser ophthalmoscope, RNFL thickness map, and texture-based en face scans or images centered on the ONH. It should be noted that only a small region of the nasal macula is included in these custom wide-field ONH scans. ONH indicates optic nerve head; RNFL, retinal nerve fiber layer; SLO, scanning laser ophthalmoscope. Figure 1 can be viewed in color online at www.glaucomajournal.com.

was never part of its own training set and was tested only once. An augmentation procedure in the form of horizontal mirroring similar to that used by Christopher et al¹² was applied to the minority class to balance the data.

Statistical Analyses

Descriptive statistics included the calculation of the mean and SD for normally distributed variables and median, first quartile, and third quartile values for non-normally distributed variables. Student *t* tests or Mann-Whitney tests were used to evaluate the statistical significance of differences in demographic and clinical parameters between glaucoma patients and healthy individuals.

Because of the age difference between healthy individuals and those with glaucoma, we conducted an age-matched subanalysis in which the age difference between healthy eyes [*n* = 54, mean age = 55.5 years (53.8, 57.9)] and glaucoma eyes [*n* = 105, mean age = 58.3 years (55.9, 60.1)] was not significant (*P* = 0.09). In addition, to evaluate the diagnostic accuracy of a multimodal DL classifier in a subset of highly myopic eyes, we used the nonhighly myopic eyes with AL ≤ 26 mm to train the DL models and the high axial myopic eyes with AL > 26 mm as a testing group. Finally, to assess the impact of astigmatism on the model performance, we conducted a subanalysis on eyes with astigmatism in healthy (*n* = 80) and glaucoma (*n* = 374) eyes. Astigmatism was defined as a cylinder error of ≥ 0.50 D. Low to moderate astigmatism was defined as a cylinder error of ≥ 0.50 D and < 3.00 D and high astigmatism as ≥ 3.00 D.³²

AUROC were used to evaluate the diagnostic accuracies of the different models investigated in the study. An AUROC regression model with a maximum likelihood estimator was applied to adjust for potentially confounding effects of age, image quality, AL, and pseudophakia.³³

Statistical analyses were performed using Stata 14.2 (StataCorp LLC). *P* values ≥ 0.05 were considered statistically significant.

RESULTS

This cross-sectional study included 463 POAG eyes of 249 patients and 130 healthy eyes of 68 individuals. Table 1 displays the demographic and clinical characteristics of the

participants and eyes. The mean (95% CI) participant age at the SD-OCT imaging date was 52.1 (50.1, 54.7) years for patients with glaucoma and 71.3 (69.3, 72.3) years for healthy individuals. POAG eyes had worse VF MD [3.78 (−4.13, −3.07)] than healthy eyes [−0.78 (−1.02, −0.34)] (*P* < 0.001). The mean spherical equivalent was significantly lower in the healthy group −3.28 (−3.11, −2.88) compared with the POAG group −1.58 (−1.98, −1.11) (*P* < 0.01). There was no significant difference in AL (*P* = 0.11) and IOP (*P* = 0.43).

AUROC and sensitivities at 80% and 95% specificity for the single-modal and multimodal DL algorithms are provided in Table 2. The multimodal DL model had a significantly higher AUROC than single-modal RNFL thickness (0.91 vs 0.84, respectively; *P* = 0.004), single-modal texture based en face images thickness (0.91 vs 0.83, respectively; *P* = 0.001), and single-modal SLO images (0.91 vs 0.68, respectively; *P* < 0.001). Moreover, the multimodal DL model using texture-based en face image thickness, RNFL thickness, and SLO image information had a significantly higher AUROC than a multimodal model using RNFL thickness and SLO images (0.91 vs 0.85, respectively; *P* = 0.02) and a similar AUROC to a multimodal model using texture and RNFL (0.91 vs 0.89, respectively; *P* = 0.21). These results indicate that adding texture information to multimodal DL models improves the classification of healthy and glaucomatous eyes.

Table 3 reports area under the curves and sensitivities at 80% and 95% specificity for all DL algorithms in the age-matched subanalysis. In this subset, the multimodal DL model had a significantly higher age-adjusted AUROC than single-modal RNFL thickness (0.91 vs 0.82; *P* = 0.001), single-modal texture based en face images thickness (0.91 vs 0.80; *P* < 0.01), and single-modal SLO images (0.91 vs 0.69; *P* < 0.001) models.

Regarding the highly myopic subgroup secondary analysis, Table 4 displays the demographic and clinical characteristics of the high axial myopia participants and eyes, and Table 5 reports AUROC and sensitivities at 80% and 95% specificity for all DL algorithms. In this high axial myopia subset, the multimodal DL model showed significantly higher diagnostic accuracy (glaucoma/no glaucoma) [0.89 (0.86, 0.92)] than the single texture-based en face image [0.83 (0.78,

TABLE 1. Patient and Eye Characteristics by Diagnosis

	Diagnosis		<i>P</i>
	Healthy (<i>n</i> = 68 persons, 130 eyes)	Glaucoma (<i>n</i> = 249 persons, 463 eyes)	
Age (y)	52.1 (50.1, 54.7)	71.3 (69.3, 72.3)	<0.001
Sex (F); %	62.1	53.4	0.01
Race (%)			
Non-White	59.2	42.5	0.003
White	40.8	57.5	—
MD (dB)	−0.78 (−1.02, −0.34)	−3.78 (−4.13, −3.07)	<0.001
IOP (mm Hg)	15.59 (13.9, 16.4)	14.3 (13.9, 15.4)	0.43
AL (mm)	24.95 (24.19, 25.3)	24.63 (24.18, 24.78)	0.11
Spherical equivalent (diopter)	−3.28 (−3.11, −2.88)	−1.58 (−1.98, −1.11)	<0.01
Cylinder error (diopter)	0.71 (0.57, 0.84)	0.95 (0.87, 1.03)	0.003
Astigmatism (%)	61.5	74.9	0.008
High astigmatism (%)	4.6	4.1	0.34

Mean values and 95% CIs are shown for continuous variables.

Statistical significance of differences in continuous and categorical variables are determined by 2-sample *t* tests and Fisher exact tests for patient-level variables (respectively) and linear mixed effects models for eye-level variables.

AL indicates axial length; IOP, intraocular pressure; MD, mean deviation.

TABLE 2. Estimated AUROC and Sensitivities at Fixed Specificities for All DL Analyses

	AUROC (95% CI)	80% specificity (%)	95% specificity (%)	Adjusted AUROC <i>P</i> value compared with multimodal model
Single-modal VGG-16 DL				
Texture-based en face image	0.83 (0.79, 0.86)	71.3 (66.2,77.1)	61.2 (57.1,66.4)	0.001
RNFL thickness map	0.84 (0.81, 0.87)	72.2 (66.9,77.5)	62.4 (59.5,65.7)	0.004
SLO image	0.68 (0.61, 0.74)	54.9 (51.7,58.3)	46.5 (42.1,47.1)	<0.001
Multimodal VGG-16 DL				
Texture and RNFL thickness	0.89 (0.86, 0.92)	82.1 (77.6,86.5)	69.8 (66.7,76.1)	0.21
RNFL thickness and SLO images	0.85 (0.84, 0.87)	73.2 (67.35,78.1)	63.9 (60.1,67.1)	0.02
Texture and SLO images	0.85 (0.83,0.87)	73.6 (68.9,79.2)	62.9 (58.3,68.1)	0.009
Texture, RNFL thickness, and SLO images	0.91 (0.87, 0.95)	84.9 (79.5,88.2)	71.1 (68.4,78.3)	—

All eyes (*n* = 593) included.

AUROC indicates area under the receiver operating curve; DL, deep learning; RNFL, retinal nerve fiber layer; SLO, scanning laser ophthalmoscope.

0.85), *P* = 0.004], RNFL thickness map [0.85 (0.81, 0.86), *P* = 0.02], and SLO image [0.69 (0.63, 0.76) *P* < 0.001] models with higher sensitivities at 80% and 95% specificities.

In the astigmatism subset, the multimodal DL model had a significantly higher age-adjusted AUROC than the single-modal RNFL thickness map (0.89 vs 0.83; *P* = 0.004), single-modal texture based en face image thickness (0.89 vs 0.77; *P* < 0.01), and single-modal SLO images (0.89 vs 0.64; *P* < 0.001) models. While the multimodal DL model had a slightly lower area under the curve when applied to eyes with astigmatism compared with all eyes (0.89 vs 0.91), it maintained its better performance compared with single-modal models. Astigmatism was defined as a cylinder error of ≥ 0.50 D. Low to moderate astigmatism was defined as a cylinder error of ≥ 0.50 D and < 3.00 D and high astigmatism as ≥ 3.00 D.³² It should be noted that our overall population sample included very few eyes with high astigmatism (ie, < 5%) defined as cylinder error ≥ 3.00 D (Table 1).

DISCUSSION

The current study compared single-modal VGG-16 CNN models trained and tested on wide-field texture-based en face images, RNFL thickness maps, and SLO images to a combined multimodal model that captured the ONH and the nasal macula trained and tested for differentiating between healthy and glaucoma eyes, regardless of AL. Results indicated that the multimodal model was

significantly better at this task than each single-modal model in eyes with and without high axial myopia. These results show promise for combining ONH measurements from multiple image types, including en face texture images.

To the best of our knowledge, this is the first study to investigate multimodal CNN-based image analysis, including information from wide-field en face OCT images. In the current sample, the sensitivity at 95% specificity of the investigated multimodal CNN was higher than the sensitivity of single-modal models.

Similar to the current study, other recent studies have investigated hybrid multimodal analyses for detecting glaucoma. It is challenging to compare the diagnostic accuracy among studies as the severity of glaucoma varies greatly among the various study populations. One can, however, compare the relative improvement in multimodal versus single-modal models. Xiong et al²⁴ developed and validated a multimodal CNN that combined VF and peripapillary OCT measurements. Results reported indicated that the tested multimodal CNN (called FusionNet) outperformed 2 glaucoma specialists at discriminating between healthy and glaucoma eyes (defined by 4 glaucoma specialists based on clinical history, VF, and OCT images).

Mehta et al³⁴ assessed the performance of an ensemble machine-learning glaucoma detection model composed of a multimodal retinal image-trained DL model (trained on OCT macula volumes and fundus photographs) combined with demographic and clinical data. This ensemble model yielded an AUROC of 0.97 for discriminating between

TABLE 3. Estimated AUROC and Sensitivities at Fixed Specificities for All DL Analyses Using Age-matched Test Set of Healthy (*n* = 54) and Glaucoma (*n* = 105) eyes

	AUROC (95% CI)	80% specificity (%)	95% specificity (%)	Adjusted AUROC <i>P</i> value compared with multimodal model
Single-modal VGG-16 DL				
Texture-based en face image	0.80 (0.78, 0.83)	68.2 (65.1,75.2)	60.1 (58.2,66.4)	<0.01
RNFL thickness map	0.82 (0.79, 0.85)	71.1 (66.9,77.5)	61.2 (57.4,66.3)	0.001
SLO image	0.69 (0.62, 0.76)	55.3 (52.1,59.9)	47.9 (43.1,49.2)	<0.001
Multimodal VGG-16 DL				
Texture and RNFL thickness	0.90 (0.86, 0.93)	83.2 (78.4,86.2)	69.3 (65.4,73.2)	0.32
RNFL thickness and SLO images	0.81 (0.79, 0.84)	69.5 (66.2,77.0)	62.4 (59.6,68.4)	0.003
Texture and SLO images	0.83 (0.79, 0.86)	72.16 (67.4,78.1)	62.9 (56.8,67.6)	0.006
Texture, RNFL thickness, and SLO images	0.91 (0.87, 0.95)	84.1 (79.3,87.9)	70.5 (67.8,74.3)	—

AUROC indicates area under the receiver operating curve; DL, deep learning; RNFL, retinal nerve fiber layer; SLO, scanning laser ophthalmoscope.

TABLE 4. Patient and Eye Characteristics by Diagnosis of High Axial Myopic Eyes

	Diagnosis		<i>P</i>
	Healthy (n = 24 persons, 44 eyes)	Glaucoma (n = 55 persons, 92 eyes)	
Age (y)	48.9 (46.7, 54.3)	65.9 (63.2, 68.5)	<0.001
Sex (F); %	54.1	43.6	0.01
Race (%)			
Non-White	67.0	42.0	0.003
White	33.0	58.0	—
MD (dB)	−1.42 (−1.89, 0.54)	−5.66 (−6.18, −4.17)	<0.001
IOP (mm Hg)	15.39 (13.8, 16.1)	14.8 (14.1, 15.3)	0.52
AL (mm)	27.0 (26.2, 27.3)	27.1 (26.5, 27.3)	0.41
Spherical equivalent (diopter)	−7.35 (−7.67, −6.98)	−4.76 (−5.21, −4.31)	0.02
History of cataract surgery	13.7	42.4	<0.001
Cylinder error (diopter)	0.83 (0.51, 0.95)	0.87 (0.43, 1.02)	0.21
Astigmatism (%)	79.45	77.8	0.19
High astigmatism (%)	2.9	3.3	0.54

Mean values and 95% CIs are shown for continuous variables. Statistical significance of differences in continuous and categorical variables are determined by 2-sample *t* tests and Fisher exact tests for patient-level variables (respectively) and linear mixed effects models for eye-level variables.

AL indicates axial length; IOP, intraocular pressure; MD, mean deviation.

healthy and glaucomatous individuals. Our results and results from the previously described studies provide compelling evidence for the usefulness of multimodal analyses in glaucoma detection.

Results from the current study highlight the value of DL analyses in glaucoma detection. Other studies have investigated the use of DL CNN analysis of OCT thickness measurements and OCT images. For instance, Asaoka et al³⁵ paired a DL CNN with transfer learning and showed an AUROC of 0.937 for discriminating between early POAG eyes (with MD > −5.0 dB) and healthy eyes, and was significantly better than Random Forest and support vector machine analyses of the same test set (AUROCs of 0.820 and 0.674, respectively), indicating the superior performance of pretrained DL models.

We also showed that combining multiple imaging modalities into a single model can aid in overcoming the challenges of optic disc imaging interpretation in highly myopic eyes described in the introduction. Other measurements that have reported better sensitivity for detecting glaucoma than circumpapillary RNFL thickness in

myopic eyes include measurement of the Bruch's membrane opening minimum rim width and the Bruch's membrane opening-based 3-dimensional neuroretinal rim thickness.^{36–39}

The high accuracy observed using RNFL thickness maps in highly myopic eyes in the current study could be attributed in part to the study population. Moreover, one of the challenges we had with myopic eyes was the high failure rate of the built-in software to segment both ILM and RNFL. This was a reason for developing the SALSA Deep software which was trained on manually segmented images with different ALs leading to an accurate segmentation of the RNFL. Rezapour et al²⁸ showed that the proposed en face texture-based images can improve on most investigated macular tissue thickness measurements for discriminating between highly myopic glaucomatous and highly myopic healthy eyes. This likely is attributable in part to its reliance on minimal tissue segmentation (segmentation of the ILM only) because attempts at multilayer segmentation tend to fail more frequently in highly myopic eyes. In addition, because vessels usually appear as shadows in OCT images, our texture-based approach assigns different values to vessels and neural tissue, whereas circumpapillary retinal nerve fiber layer images can include both non-neural tissue (ie, vessels) and neural tissue in resulting thickness measurements.

There are limitations to the current study. First, although 5-fold cross-validation with independent test and training sets was used to control for overfitting of models, multimodal and single-modal CNN models were not tested on external test data sets so accuracy may be reduced in other populations. In addition, our sample size was relatively small for CNN modeling, although the sample size was large enough to detect significant differences among analysis strategies. It should be noted that the sample size of the highly myopic eyes was too small to include in the training set; results may improve when highly myopic eyes are included in the training. Finally, the current results cannot be generalized to highly myopic eyes with significantly high astigmatism because our sample contained few such eyes. Future directions to improve classification results of the current OCT-based CNN model could include adding VF measurements to the model similar to recent work by Xiong et al.²⁴ In addition, it would be important to determine whether the results are generalized to other OCT instruments in external data sets. Assessment of model performance in more highly myopic eyes with high astigmatism also is important for generalizability.

TABLE 5. Estimated AUROC and Sensitivities at Fixed Specificities for all DL Analyses in High Axial Myopic Eyes

	AUROC (95% CI)	80% specificity (%)	95% specificity (%)	Adjusted AUROC <i>P</i> value compared with multimodal model
Single-modal VGG-16 DL				
Texture-based en face image	0.83 (0.78, 0.85)	71.3 (66.5,76.4)	63.5 (59.4,64.6)	0.004
RNFL thickness map	0.85 (0.81, 0.86)	73.0 (68.2,78.4)	61.2 (60.1,65.3)	0.02
SLO image	0.69 (0.63, 0.76)	56.1 (53.2,60.1)	48.3 (44.1,50.3)	<0.001
Multimodal VGG-16 DL				
Texture and RNFL thickness	0.87 (0.85, 0.90)	83.2 (78.4,86.3)	68.9 (66.3,74.2)	0.42
RNFL thickness and SLO images	0.84 (0.79, 0.85)	72.1 (67.3,77.6)	64.5 (60.4, 65.7)	0.01
Texture and SLO images	0.85 (0.82, 0.86)	74.2 (69.9,80.7)	62.5 (61.4,66.6)	0.03
Texture, RNFL thickness, and SLO images	0.89 (0.86, 0.92)	85.1 (80.2,89.3)	71.2 (66.8,76.4)	—

AUROC indicates area under the receiver operating curve; DL, deep learning; RNFL, retinal nerve fiber layer; SLO, scanning laser ophthalmoscope.

CONCLUSION

The current results indicate that combining CNN models trained and tested on currently available OCT RNFL and SLO images with a novel en face texture-based CNN model increases healthy versus glaucoma classification success in all eyes, including those that are highly myopic. This finding is important because the similar performance of the proposed model in highly myopic eyes allows acceptable healthy versus glaucoma classification from ONH imaging without additional macular imaging. Future work should aid in developing DL analyses for clinical use.

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